

TrustSphere

Trust Every Session, Not Just Every Login

Privacy-First

Risk-Based

Continuous Auth

RBI Compliant



Problem & Solution

The Identity Trust Gap



Static Auth Is Blind

Login-only checks miss fraud that starts mid-session



Rule Engines Cannot Adapt

Fixed thresholds are outsmarted by evolving attacks



Insider Risk Is Invisible

Employee privilege misuse evades detection for days



✓ TrustSphere Approach



Continuous Session Scoring

Real-time 0-100 risk score on every event, not just login



Risk-Triggered Step-Up Auth

Zero friction for safe users — escalates only on elevated risk



ML Anomaly Detection

IsolationForest + LightGBM catch what rules and humans miss

Seamless for trusted users · Challenged only when risk is elevated · Privacy preserved throughout

Core Capabilities

Six independent detection engines forming a unified, continuously updated trust score

1



Continuous Risk Engine

6-signal weighted score · real-time

2



Behavioural Biometrics

Keystroke · scroll · touch patterns

3



Device Intelligence

Fingerprint · trust level · emulator flags

4



KYC Fraud Scoring

Liveness · docs · watchlist · synthetic ID

5



Privileged Access Monitor

Bulk access · off-hours · export anomalies

6

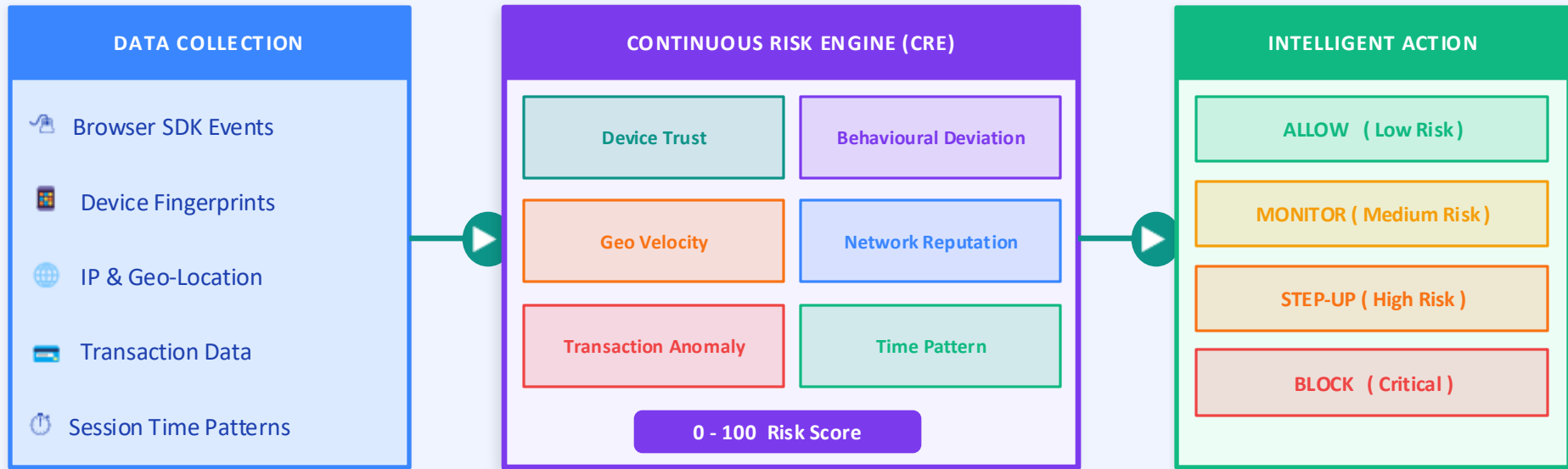


Step-Up Authentication

Push · OTP · biometric · video KYC

Technical Architecture

End-to-end signal flow — collection → scoring → intelligent action



All ML runs on CPU only · 100% open-source · No paid APIs · No vendor lock-in

Feasibility & Viability

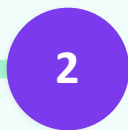
Real infrastructure · Production-grade · Deployable today



Phase 1

Core Engine

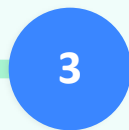
Days 1-4



Phase 2

ML & Biometrics

Days 5-8



Phase 3

Integration

Days 9-12



Phase 4

Scale & Compliance

Days 13-16



100% Open Source

No paid APIs or vendor lock-in



CPU-Only ML

No GPU needed for any inference



REST API First

Integrates with any core banking system



Multi-Tenant Ready

Isolated per-bank data and policies



RBI Framework Aligned

One-click compliance report generation



Scales with Volume

Celery + Redis handle load bursts

Impact & Benefits

↓ 90%

Account Takeover
Risk Reduction

↓ 80%

KYC Fraud
Incidents Prevented

0

Friction Added
For Safe Users

✓ RBI

Compliance
Reporting Automated

Who Benefits



Banks & NBFCs

Reduced fraud losses +
RBI alignment



End Customers

Seamless, secure
banking experience



Security Analysts

ML-triaged alerts, clear
dashboards



Compliance Teams

Automated RBI reports
on demand

Every Session Trusted. Every Attack Stopped. Every User Protected.